

1) Review

With linear regression we minimized ~~sum~~^{mean} squared error

$$\text{minimize } J(\theta) = \frac{1}{2m} \sum_{i=1}^m \underbrace{(h_{\theta}(x^{(i)}) - y^{(i)})^2}_{\text{loss function}}$$

cost function

A loss function in this case measures the error for a single ^{data}_(sample) point. The cost function~~s~~ measures some statistic across a set of samples.

With ^{binary} classification, every classification is either right or wrong. If there are more than two choices (classes) then a classification is either right or a match to one of the incorrect classes. We can minimize error by always picking the class with the highest probability. The probability of error thus becomes the probability that the ~~classifi~~^{true} class is any other class than the ~~the~~ class with the highest probability. When there are two classes this becomes

$$\text{minimize } P(\text{error}) = \int_{\bar{x} \in \mathbb{R}^n} \min(P(w_1 | \bar{x}), P(w_2 | \bar{x})) p(\bar{x}) d\bar{x}$$

If we consider the above a cost function, it is a cost function that error has equivalent to ~~loss~~ cost.

What if cost $\neq$ error?

Sometimes cost is higher ~~if~~ ^{if} we depending on the type of error. This is why we sometimes bias a classifier toward precision or recall. We presented examples in the last lecture.

To deal with the case where not all ^{types} events result in the same loss, we need to introduce some new concepts.

$$\lambda_{ij} \triangleq \lambda(\alpha_i | \omega_j) = \text{the loss derived from taking action } \alpha_i$$

What are actions?

Detecting cancer is a decision.

The action might be chemotherapy.

Chemotherapy introduces real health risks.

What is risk?

Risk = the loss we expect, on average, from taking an action. We don't know if the target is friend or foe, so we weigh each loss by how likely each one is.

Example: drone strike

A drone classifies targets as "friend" or "foe."

$\omega_1 = \text{friend}$, $\omega_2 = \text{foe}$

$\alpha_1 = \text{spare the target (do nothing)}$, $\alpha_2 = \text{eliminate the target}$

$$\lambda_{11} = \lambda(\alpha_1 | \omega_1) = 0 \quad \leftarrow \text{spare a friend}$$

$$\lambda_{12} = \lambda(\alpha_1 | \omega_2) = 1 \quad \leftarrow \text{spare a foe}$$

$$\lambda_{21} = \lambda(\alpha_2 | \omega_1) = 100 \quad \leftarrow \text{eliminate a friend}$$

$$\lambda_{22} = \lambda(\alpha_2 | \omega_2) = 0 \quad \leftarrow \text{eliminate a foe}$$

We place a much higher loss on eliminating a friend than on sparing a foe.

The choice of losses here is arbitrary, but we must assign some finite loss on eliminating a friend or the resulting system would spare everyone.

$R(\alpha) =$ expected value of the loss function
conditional risk:

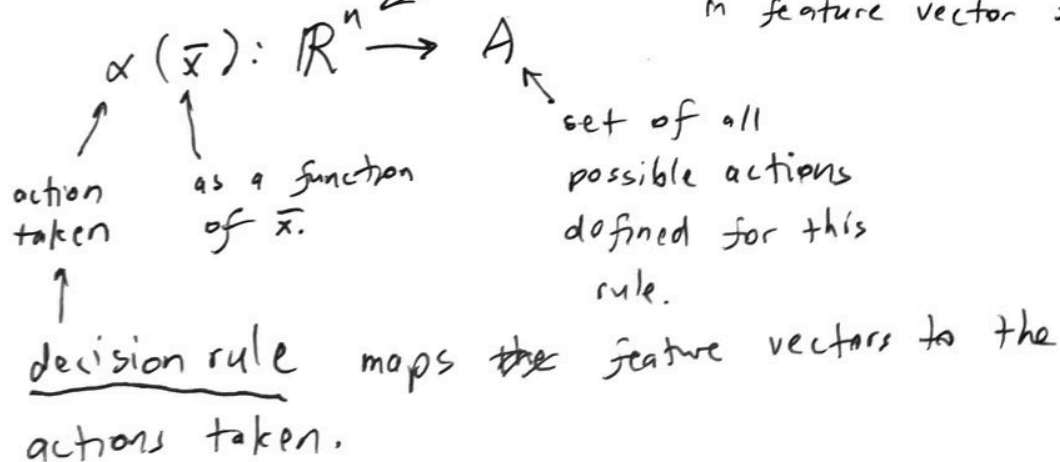
$$R(\alpha_i | \bar{x}) = \sum_{j=1}^k \lambda(\alpha_i | \omega_j) P(\omega_j | \bar{x}) \quad \text{where } k = \# \text{ of classes} \quad (1)$$

The conditional risk is the expected loss given feature vector $\bar{x}$.

How we assign losses to each action allows us to bias the decision rule. A higher loss for an action will bias decisions away from that action.

Given the objective of minimizing risk, what decision rule should we employ?

All decision rules follow $\alpha: \mathbb{R}^n \rightarrow A$ $n = \#$ of features (dimensions)
 n feature vector x .



Bayes' Risk

$$R(\alpha) = \int_{\bar{x} \in \mathbb{R}^n} R(\alpha(\bar{x}) | \bar{x}) p(\bar{x}) d\bar{x} \quad (2)$$

Bayes' Risk is the expected loss (risk) for a given decision rule α .

Bayes' Risk (cont.)

α^* denotes the decision rule that minimizes Bayes' risk and is called Bayes' Decision Rule.

$$R(\alpha^*) = \int_{\bar{x} \in \mathbb{R}^n} \left[\min_{i=1, \dots, M} R(\alpha_i | \bar{x}) \right] p(\bar{x}) d\bar{x}$$

of actions

~~Any~~ Any given decision α_i that increases $R(\alpha_i | \bar{x})$ the integral would increase. That isn't a proof but just ponder it a bit. Is there any way that choosing the riskier action given $\bar{x}$ could decrease the expected risk?

More formally, for any decision rule α

$$R(\alpha) = \int_{\bar{x} \in \mathbb{R}^n} R(\alpha | \bar{x}) p(\bar{x}) d\bar{x} \geq \int_{\bar{x} \in \mathbb{R}^n} \left[\min_{i=1, \dots, M} R(\alpha_i | \bar{x}) \right] p(\bar{x}) d\bar{x}$$

i.e.,

$$R(\alpha) \geq R(\alpha^*)$$

Example 1: Binary classification

$$\lambda_{ij} = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} \text{ where } \lambda_{12} > \lambda_{22} \text{ and } \lambda_{21} > \lambda_{11}$$

(loss) taking α_2 when class is ω_1

Use Bayes' decision rule.

decide $\begin{cases} \alpha_1 & \text{if } R(\alpha_1 | \bar{x}) < R(\alpha_2 | \bar{x}) \\ \alpha_2 & \text{otherwise.} \end{cases}$

loss of a mistake

loss of a correct action

Example 1 (cont.)

Using the definition of the conditional risk (1)

(4)

we can rewrite (3) as

$$\text{decide } \begin{cases} \omega_1 & \text{if } \frac{\lambda_{11} p(\bar{x}|\omega_1) P(\omega_1) + \lambda_{12} p(\bar{x}|\omega_2) P(\omega_2)}{p(\bar{x})} \\ & < \frac{\lambda_{21} p(\bar{x}|\omega_1) P(\omega_1) + \lambda_{22} p(\bar{x}|\omega_2) P(\omega_2)}{p(\bar{x})} \\ \omega_2 & \text{otherwise} \end{cases}$$

The probability $p(\bar{x})$ is the same on both sides of the inequality in (4), so it cancels out leaving

(5)

$$\text{Decide } \begin{cases} \omega_1 & \text{if } \lambda_{11} p(\bar{x}|\omega_1) P(\omega_1) + \lambda_{12} p(\bar{x}|\omega_2) P(\omega_2) \\ & < \lambda_{21} p(\bar{x}|\omega_1) P(\omega_1) + \lambda_{22} p(\bar{x}|\omega_2) P(\omega_2) \\ \omega_2 & \text{otherwise} \end{cases}$$

Rearranging (5) yields

(6)

$$\text{Decide } \begin{cases} \omega_1 & \text{if } (\lambda_{11} - \lambda_{21}) p(\bar{x}|\omega_1) P(\omega_1) < (\lambda_{22} - \lambda_{12}) p(\bar{x}|\omega_2) P(\omega_2) \\ \omega_2 & \text{otherwise} \end{cases}$$

Likelihood ratios

A "likelihood ratio" is the ratio of the probability of one observation (a sample) given it came from one class and for that same observation coming from another class

$$\left. \frac{p(\bar{x}|\omega_1)}{p(\bar{x}|\omega_2)} \right\} \text{likelihood ratio}$$

(7)

Likelihood ratios

Let's continue example 1 using likelihood ratios

From (6) and (7)

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{p(\bar{x}|\omega_1)}{p(\bar{x}|\omega_2)} > \frac{\lambda_{22}-\lambda_{12}}{\lambda_{11}-\lambda_{21}} \frac{P(\omega_2)}{P(\omega_1)} \\ \omega_2 & \text{otherwise} \end{cases} \quad (8)$$

comparison
flipped because

$$\lambda_{22} < \lambda_{12}$$

$$\text{thus } \lambda_{11} - \lambda_{21} < 0$$

we divided both sides by
a negative number.

Likelihood Ratios with Priors

move the prior probability ratio from the right hand side of
(8) to the left hand side

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{p(\bar{x}|\omega_1) P(\omega_1)}{p(\bar{x}|\omega_2) P(\omega_2)} > \frac{\lambda_{22}-\lambda_{12}}{\lambda_{11}-\lambda_{21}} \\ \omega_2 & \text{otherwise} \end{cases} \quad (9)$$

Likelihood Ratios w/ Priors (cont.) for binary classification

How would we define a loss function that minimizes error (i.e., maximizes accuracy)?

- 1) we would expect zero loss for obtaining the correct answer.

$$\lambda_{11} = \lambda_{22} = 0$$

- 2) we would expect the same loss for making either incorrect answer.

$$\lambda_{12} = \lambda_{21} = 1$$

With this ~~loss~~ assignment the optimal classifier uses the decision rule

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{p(\bar{x}|\omega_1)P(\omega_1)}{p(\bar{x}|\omega_2)P(\omega_2)} > \frac{0-1}{0-1} \\ \omega_2 & \text{otherwise} \end{cases}$$

which becomes

$$\text{Declare } \begin{cases} \omega_1 & \text{if } \frac{p(\bar{x}|\omega_1)P(\omega_1)}{p(\bar{x}|\omega_2)P(\omega_2)} > 1 \\ \omega_2 & \text{otherwise} \end{cases} \quad (10)$$

Likelihood Ratios w/ Priors (cont.)

In practice, we should avoid ratios as a decision rule for at least the following reasons

1) Numerical stability

Likelihoods are products of many numbers, one for each feature or sample.

Probabilities are at most 1, so multiplying many of them shrinks the product.

Density values can exceed 1 (only the integral over a range must be at most 1), so a product of densities can shrink or grow.

Either way, a long product can underflow or overflow in floating point.

A ratio of two such products can then be $0/0$ or ∞/∞ .

2) Difficult to interpret ~~due to large dynamic range~~ Values with a wide range in the magnitude are difficult to visualize. Often times when dealing with values that may have ~~instability~~ wildly different order of magnitude, we use log plots.

A common alternate way of dealing with likelihoods is to use log of the likelihoods.

Additional benefits of log likelihoods

1) Easier math

In ML, likelihoods are products of probabilities, a log of a product converts a product into a sum.

representation 1
 ab

representation 2

$$\log(ab) = \log a + \log b$$

1) Easier math (cont.)

Using ~~log~~ sum of logs simplifies taking derivatives and thus simplifies gradient descent. Calculating the gradient of the log likelihood is easier.

Log Likelihood Ratios

$$\log \left(\frac{p(\bar{x}|w_1) P(w_1)}{p(\bar{x}|w_2) P(w_2)} \right) = \log p(\bar{x}|w_1) + \log P(w_1) - \log p(\bar{x}|w_2) - \log P(w_2)$$

when minimizing error

The decision ~~function~~ rule becomes

$$\begin{aligned} &\text{declare } w_1 \quad \text{if } \log p(\bar{x}|w_1) + \log P(w_1) - \log p(\bar{x}|w_2) - \log P(w_2) > 0 \\ &w_2 \quad \text{otherwise} \end{aligned} \quad (11)$$

When we have a loss function λ_{ij} such that

$$\lambda = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} \quad \text{where } \begin{cases} \lambda_{12} > \lambda_{22} \\ \lambda_{21} > \lambda_{11} \end{cases} \quad \left. \begin{array}{l} \text{loss of taking} \\ \text{the wrong action} \\ \text{is greater than the} \\ \text{loss of the correct} \\ \text{action} \end{array} \right\}$$

the ~~log~~ likelihood ratio w/ priors was given by (9).

(9) can be restated as a loss likelihood ratio w/ priors as

$$\text{declare } \begin{cases} w_1 & \text{if } \log \left(\frac{p(\bar{x}|w_1)}{p(\bar{x}|w_2)} \cdot \frac{P(w_1)}{P(w_2)} \right) > \log \left(\frac{\lambda_{22} - \lambda_{12}}{\lambda_{11} - \lambda_{21}} \right) \\ w_2 & \text{otherwise} \end{cases} \quad (12)$$

to make the denominator and numerator both ~~no~~ positive, multiply the loss ratio by $\frac{1}{\lambda_{21} - \lambda_{11}}$

Rearranging (12) yields

$$\text{declare } \begin{cases} w_1 & \text{if } \log p(\bar{x}|w_1) + \log P(w_1) - \log p(\bar{x}|w_2) - \log P(w_2) \\ & > \log(\lambda_{12} - \lambda_{22}) - \log(\lambda_{21} - \lambda_{11}) \\ w_2 & \text{otherwise} \end{cases}$$

Moving all logs to the lefthand side of the inequality yields

$$\text{declare } \begin{cases} w_1 & \log p(\bar{x}|w_1) + \log P(w_1) - \log p(\bar{x}|w_2) \\ & - \log P(w_2) - \log(\lambda_{12} - \lambda_{22}) + \log(\lambda_{21} - \lambda_{11}) > 0 \\ w_2 & \text{otherwise} \end{cases} \quad (13)$$

(13) expresses the decision rule for a binary classifier that minimizes risk for the loss function λ .

Increasing the loss incurred by a mistake means increasing λ_{12} and/or λ_{21} . In (13), increasing λ_{12} also increases $\log(\lambda_{12} - \lambda_{22})$ which requires larger values for $\log p(\bar{x}|w_1) + \log P(w_1)$ to overcome the loss for taking the action for deciding w_1 .

Increasing λ_{12} biases the classifier towards deciding w_2 .

If w_2 means "person" and w_1 means not person, the system would be biased toward identifying a person.

